

Automated Vehicles as a Space for Work & Wellbeing

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ABSTRACT

The objective of this in-person CHI course is to provide CHI attendees with an introduction and overview of the rapidly evolving field of automotive user interfaces (AutomotiveUI). The course will focus on UI aspects in the transition towards automated driving. In particular, we will also discuss the opportunities of cars as a new space for non-driving-related activities, such as work, relaxation, and play. For newcomers and experts of other HCI fields, we will present the special properties of this field of HCI and provide an overview of new opportunities, but also general design and evaluation aspects of novel automotive user interfaces.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; Ubiquitous and mobile computing.

KEYWORDS

Automotive user interfaces; non-driving-related activities; manual and automated driving; work and wellbeing.

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1 INTRODUCTION

In recent years, driving a car became more than the activity of maneuvering the car to a destination. In addition to the primary driving task, drivers also want to perform non-driving-related tasks [19, 25] including the operation of safety and comfort functions. Also, drivers want to be entertained (e.g., listen to music) or communicate while on the go. With the release of vehicles in different levels of automation, we expect this demand to become even more relevant and pressing. This opens the opportunity for novel activities in the car, such as work, play, and relaxation.

Already today, drivers use mobile devices in the car for these tasks. While driving manually, this can distract them from their primary driving task, posing humans at risk. Thus, the challenge for researchers and developers is to enable such non-driving-related

tasks in the car in the transition to automated cars, while ensuring driving safety and comfort.

In contrast to our first courses at CHI, at MobileHCI, and MUM [20, 21, 23], we will now especially focus on the transition to automated driving [10, 11] and how this makes the car an interesting and novel area for HCI research and applications, e.g., related to work, play, and relaxation. We provide an introduction to automotive UIs and highlights the special properties of this HCI sub-field. By outlining the requirements, the design, and evaluation of automotive UIs, we discuss the specific requirements for in-car applications.

2 BENEFITS

The objective of this course is to provide newcomers to the field of AutomotiveUI with an introduction and overview of the field and show how it is different from others. An intensive discussion of the requirements, trends, and challenges will help participants to get an overview and might be the basis for new research ideas. Especially with automated driving, we see the car as a platform that becomes more and more important for interactive applications and, thus, an important focus for human-computer interaction practice and research. We want attendees to leave with a fresh perspective on the field and inspire them to build the next generation of in-vehicle interfaces or services.

3 INTENDED AUDIENCE

We target a broad audience including AutomotiveUI novices (students, industrial / academic researchers), but also researchers, practitioners, & designers with experiences in creating AutomotiveUIs. With driving automation, we see the car as a novel platform for interactive systems, which is interesting for attendees of a variety of backgrounds.

4 PREREQUISITES

The expected audience should have a basic knowledge of HCI. This could be a previously attended course or a basic university lecture or experiences from prior projects in this domain. There are no additional prerequisites.

5 CONTENT & PRACTICAL WORK

The objective of this CHI course is to provide attendees with an introduction and overview of the rapidly evolving field of automotive user interfaces (AutomotiveUI). The course will focus on UI aspects in the transition towards automated driving. In particular, we will discuss the opportunities of cars as a new space for non-driving-related activities, such as work, relaxation, and play.

We will begin with an overall introduction to AutomotiveUI. This includes a discussion of important terms, such as driving

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task, driver distraction, vehicle systems, and automation levels, as well as current challenges and expected trends with regard to AutomotiveUI and the more general topic of human mobility.

In the second part of the course, we will provide an overview of guidelines and requirements for the design of in-car systems, including principles, guidelines, and standards. Our discussion of non-driving related tasks within the context of cars and human mobility, will be followed by an interactive hands-on design exercise.

In the last part of the course, participants will learn from a panel of experts, how automotive applications and interfaces can be evaluated during design. This includes the explanation of various evaluation measures, different testing procedures (e.g., Lane Change Task, Detection Response Task, and field trials), and platforms.

6 PRESENTATION FORMAT & COURSE ADVERTISEMENT

We present this course as an interactive course with slides, videos, and group discussions and activities. We successfully used this format for AutomotiveUI courses at CHI'21, CHI'16 and '17 [20, 21], MobileHCI '17 [23], MUM '18, and Mensch & Computer '15. We also taught this course online at CHI 2021 [24]. By applying active learning methods, we strive to adapt the materials to the specific (existing) knowledge and interests of the audience.

The call for participation will be distributed via HCI, UX, and AutomotiveUI related mailing lists. We will further use our personal distribution lists and create a course website.

7 INSTRUCTOR BACKGROUNDS

Bastian Pfleging (www.bastian-pfleging.eu) is professor of Ubiquitous Computing and Smart Systems at TU Bergakademie Freiberg, Germany. His research interests are user interfaces for future mobility [29], multimodal interaction [2, 27], workload [22, 26], external interfaces for automated cars (e.g., [1, 3, 4, 6–8]), and non-driving-related activities [5, 9, 19, 25]. He was involved in co-organizing various workshops and courses (e.g., at AutomotiveUI, and CHI, [17, 18, 28]). Bastian is actively involved in organizing conferences like CHI, UIST, AutomotiveUI, MobileHCI and is steering committee member of AutomotiveUI.

Andrew L. Kun (www.andrewkun.com) is professor of Electrical and Computer Engineering at the University of New Hampshire. His research focus is human-computer interaction in vehicles [12–16], as well as the use of visual behavior and pupil diameter measures [22] to assess and improve the design of interfaces. He is a member of the steering committee of AutomotiveUI.

Orit Shaer (cs.wellesley.edu/~oshaer/) is professor of computer science and Media Arts and Sciences at Wellesley College. Her expertise is in designing, implementing and evaluating novel human-computer interactions including augmented reality (e.g. [31]), and tangible interaction [30]. She served as program co-chair of the 2020 ACM Tangible, Embedded, and Embodied Interaction (TEI) conference.

8 RESOURCES

Additional details about the course as well as accompanying material will be published on our *course website* at <https://ubisys.org/chi2022-course-future-cars/>. It provides information about the

course and links to related material, so that participants can get familiar with the scope of the subject and the goals of the course.

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REFERENCES

- [1] Sander Ackermans, Debargha Dey, Peter Ruijten, Raymond H. Cuijpers, and Bastian Pfleging. 2020. The Effects of Explicit Intention Communication, Conspicuous Sensors, and Pedestrian Attitude in Interactions with Automated Vehicles. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376197>
- [2] Michael Braun, Nora Broy, Bastian Pfleging, and Florian Alt. 2019. Visualizing natural language interaction for conversational in-vehicle information systems to minimize driver distraction. *Journal on Multimodal User Interfaces* 13, 2 (2019), 71–88. <https://doi.org/10.1007/s12193-019-00301-2>
- [3] Ashley Colley, Jonna Häkkinä, Bastian Pfleging, and Florian Alt. 2017. A Design Space for External Displays on Cars. In *Proceedings of the 9th International Conference on Automotive User Interfaces and Interactive Vehicular Applications Adjunct* (Oldenburg, Germany) (AutomotiveUI '17). Association for Computing Machinery, New York, NY, USA, 146–151. <https://doi.org/10.1145/3131726.3131760>
- [4] Ashley Colley, Jonna Häkkinä, Meri-Tuulia Forsman, Bastian Pfleging, and Florian Alt. 2018. Car Exterior Surface Displays: Exploration in a Real-World Context. In *Proceedings of the 7th ACM International Symposium on Pervasive Displays* (Munich, Germany) (PerDis '18). Association for Computing Machinery, New York, NY, USA, Article 7, 8 pages. <https://doi.org/10.1145/3205873.3205880>
- [5] Henrik Detjen, Bastian Pfleging, and Stefan Schneegass. 2020. A Wizard of Oz Field Study to Understand Non-Driving-Related Activities, Trust, and Acceptance of Automated Vehicles. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Virtual Event, DC, USA) (AutomotiveUI '20). Association for Computing Machinery, New York, NY, USA, 19–29. <https://doi.org/10.1145/3409120.3410662>
- [6] Debargha Dey, Azra Habibovic, Bastian Pfleging, Marieke Martens, and Jacques Terken. 2020. Color and Animation Preferences for a Light Band EHMI in Interactions Between Automated Vehicles and Pedestrians. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376325>
- [7] Debargha Dey, Kai Holländer, Melanie Berger, Berry Eggen, Marieke Martens, Bastian Pfleging, and Jacques Terken. 2020. Distance-Dependent EHMIs for the Interaction Between Automated Vehicles and Pedestrians. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Virtual Event, DC, USA) (AutomotiveUI '20). Association for Computing Machinery, New York, NY, USA, 192–204. <https://doi.org/10.1145/3409120.3410642>
- [8] Debargha Dey, Andrii Matvienko, Melanie Berger, Bastian Pfleging, Marieke Martens, and Jacques Terken. 2021. Communicating the intention of an automated vehicle to pedestrians: The contributions of eHMI and vehicle behavior. *it - Information Technology* 63, 2 (2021), 123–141. <https://doi.org/10.1515/itit-2020-0025>
- [9] Renate Häußelschmid, Bastian Pfleging, and Andreas Butz. 2017. The Influence of Non-driving-Related Activities on the Driver's Resources and Performance. In *Automotive User Interfaces: Creating Interactive Experiences in the Car*, Gerrit Meixner and Christian Müller (Eds.). Springer International Publishing, Cham, 215–247. https://doi.org/10.1007/978-3-319-49448-7_8
- [10] Christian P. Janssen, Linda Ng Boyle, Andrew L. Kun, Wendy Ju, and Lewis L. Chuang. 2019. A Hidden Markov Framework to Capture Human-Machine Interaction in Automated Vehicles. *International Journal of Human-Computer Interaction* 35, 11 (2019), 947–955. <https://doi.org/10.1080/10447318.2018.1561789>
- [11] Christian P. Janssen, Shamsi T. Iqbal, Andrew L. Kun, and Stella F. Donker. 2019. Interrupted by my car? Implications of interruption and interleaving research for automated vehicles. *International Journal of Human-Computer Studies* 130 (2019), 221–233. <https://doi.org/10.1016/j.ijhcs.2019.07.004>
- [12] Andrew L. Kun. 2018. Human-Machine Interaction for Vehicles: Review and Outlook. *Foundations and Trends® in Human-Computer Interaction* 11, 4 (2018), 201–293. <https://doi.org/10.1561/11000000069>
- [13] Andrew L. Kun, S. Boll, and A. Schmidt. 2016. Shifting Gears: User Interfaces in the Age of Autonomous Driving. *IEEE Pervasive Computing* 15, 1 (Jan 2016), 32–38. <https://doi.org/10.1109/MPRV.2016.14>
- [14] Andrew L. Kun, Hidde van der Meulen, and Christian P. Janssen. 2017. Calling while driving: An initial experiment with HoloLens. In *Proceedings of Driving Assessment 2017* (Manchester Village, Vermont).

- [15] Andrew L. Kun, Jerry Wachtel, W. Thomas Miller, Patrick Son, and Martin Laval-lière. 2015. User Interfaces for First Responder Vehicles: Views from Practitioners, Industry, and Academia. In *Proceedings of the 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Nottingham, United Kingdom) (*AutomotiveUI '15*). Association for Computing Machinery, New York, NY, USA, 163–170. <https://doi.org/10.1145/2799250.2799289>
- [16] Zeljko Medenica and Andrew L. Kun. 2007. Comparing the influence of two user interfaces for mobile radios on driving performance. In *Proceedings of Driving Assessment 2007* (Stevenson, Washington).
- [17] Alexander Meschtscherjakov, Manfred Tscheligi, Bastian Pflöging, Shadan Sadeghian Borojeni, Wendy Ju, Philippe Palanque, Andreas Riener, Bilge Mutlu, and Andrew L. Kun. 2018. Interacting with Autonomous Vehicles: Learning from Other Domains. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (*CHI EA '18*). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3170427.3170614>
- [18] Alexander Meschtscherjakov, Manfred Tscheligi, Dalila Szostak, Sven Krome, Bastian Pflöging, Rabindra Ratan, Ioannis Politis, Sonia Baltodano, Dave Miller, and Wendy Ju. 2016. HCI and Autonomous Vehicles: Contextual Experience Informs Design. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (*CHI EA '16*). Association for Computing Machinery, New York, NY, USA, 3542–3549. <https://doi.org/10.1145/2851581.2856489>
- [19] Bastian Pflöging. 2017. *Automotive User Interfaces for the Support of Non-Driving-Related Activities*. Ph.D. Dissertation. University of Stuttgart, Stuttgart, Germany. <https://doi.org/10.18419/opus-9090>
- [20] Bastian Pflöging, Nora Broy, and Andrew L. Kun. 2016. An Introduction to Automotive User Interfaces. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (*CHI EA '16*). Association for Computing Machinery, New York, NY, USA, 961–964. <https://doi.org/10.1145/2851581.2856685>
- [21] Bastian Pflöging, Nora Broy, and Andrew L. Kun. 2017. An Introduction to Automotive User Interfaces. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) (*CHI EA '17*). Association for Computing Machinery, New York, NY, USA, 1244–1248. <https://doi.org/10.1145/3027063.3027110>
- [22] Bastian Pflöging, Drea K. Fekety, Albrecht Schmidt, and Andrew L. Kun. 2016. A Model Relating Pupil Diameter to Mental Workload and Lighting Conditions. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (*CHI '16*). Association for Computing Machinery, New York, NY, USA, 5776–5788. <https://doi.org/10.1145/2858036.2858117>
- [23] Bastian Pflöging, Andrew L. Kun, and Nora Broy. 2017. The Car as an Environment for Mobile Devices. In *Proceedings of the 19th International Conference on Human-Computer Interaction with Mobile Devices and Services* (Vienna, Austria) (*MobileHCI '17*). Association for Computing Machinery, New York, NY, USA, Article 63, 5 pages. <https://doi.org/10.1145/3098279.3119916>
- [24] Bastian Pflöging, Andrew L. Kun, and Orit Shaer. 2021. Automated Vehicles as a Space for Work & Wellbeing. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, Article 140, 3 pages. <https://doi.org/10.1145/3411763.3445012>
- [25] Bastian Pflöging, Maurice Rang, and Nora Broy. 2016. Investigating User Needs for Non-driving-related Activities During Automated Driving. In *Proceedings of the 15th International Conference on Mobile and Ubiquitous Multimedia* (Rovaniemi, Finland) (*MUM '16*). Association for Computing Machinery, New York, NY, USA, 91–99. <https://doi.org/10.1145/3012709.3012735>
- [26] Florian Roeder, Sonja Rümelin, Bastian Pflöging, and Tom Gross. 2017. The Effects of Situational Demands on Gaze, Speech and Gesture Input in the Vehicle. In *Proceedings of the 9th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Oldenburg, Germany) (*AutomotiveUI '17*). Association for Computing Machinery, New York, NY, USA, 94–102. <https://doi.org/10.1145/3122986.3122999>
- [27] Florian Roeder, Sonja Rümelin, Bastian Pflöging, and Tom Gross. 2019. Investigating the effects of modality switches on driver distraction and interaction efficiency in the car. *Journal on Multimodal User Interfaces* 13, 2 (2019), 89–97. <https://doi.org/10.1007/s12193-019-00297-9>
- [28] Shadan Sadeghian Borojeni, Alexander Meschtscherjakov, Bastian Pflöging, Birsan Donmez, Andreas Riener, Christian P. Janssen, Andrew L. Kun, Wendy Ju, Christian Remy, and Philipp Wintersberger. 2020. Should I Stay or Should I Go? Automated Vehicles in the Age of Climate Change. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI EA '20*). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3375162>
- [29] Stefan Schneeß, Bastian Pflöging, Dagmar Kern, and Albrecht Schmidt. 2011. Support for Modeling Interaction with Automotive User Interfaces. In *Proceedings of the 3rd International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Salzburg, Austria) (*AutomotiveUI '11*). Association for Computing Machinery, New York, NY, USA, 71–78. <https://doi.org/10.1145/2381416.2381428>
- [30] Orit Shaer, Eva Hornecker, et al. 2010. Tangible User Interfaces: Past, Present, and Future Directions. *Foundations and Trends® in Human-Computer Interaction* 3, 1–2 (2010), 4–137. <https://doi.org/10.1561/1100000026>
- [31] Hidde van der Meulen, Andrew L. Kun, and Orit Shaer. 2017. What Are We Missing?: Adding Eye-Tracking to the HoloLens to Improve Gaze Estimation Accuracy. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces* (Brighton, United Kingdom) (*ISS '17*). Association for Computing Machinery, New York, NY, USA, 396–400. <https://doi.org/10.1145/3132272.3132278>